

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]

Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping Scoring Analysis

Given

- Instructed: 10 taps
- Detected: 11 taps (likely includes an extra partial/initial/final peak; we evaluate the 10-tap sequence per instruction)
- Tap periods (frames) for 10 intervals: [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]
- Tap amplitudes (pixels) for 11 peaks: [129.65, 129.54, 115.83, 110.02, 116.85, 107.34, 95.02, 116.76, 116.83, 114.13, 116.81]

Note: Periods are the inter-tap intervals (10 values), amplitudes are per tap (11 values). We consider the first 10–11 peaks as one 10-tap sequence with one extra detection; the trends are clear regardless.

Speed and rhythm (periods)

- Mean period $\approx$ 103 frames with moderate variability (range 92–123).
- No sustained slowing trend; values fluctuate but remain within a relatively tight band for finger tapping.
- No obvious long arrest/freeze; no period stands out as a clear outlier indicating a halt.

- Rhythm shows minor variability without ≥ 3 distinct interruptions. This aligns more with “slight slowing or minor hesitations” rather than moderate/severe impairment.

Amplitude pattern

- Early taps are largest: ~129.6, 129.5.
- Then a drop by tap 3–7: 115.8 → 110.0 → 116.9 → 107.3 → 95.0 (lowest).
- After that, amplitudes rebound: ~116.8, 116.8, 114.1, 116.8.

Interpretation:

- There is a clear amplitude decrement starting after the first two taps, reaching a minimum around the 7th tap, followed by partial recovery.
- According to criteria:
 - “Decrements near the end” → Slight (1).
 - “Decrements midway through the 10-tap sequence” → Mild (2).
 - “Decrements starting after the 1st tap” → Moderate (3).

Here, the decrement begins early (from tap 3 onward) and is measurable, with the lowest amplitude by mid-sequence. That best matches “midway through” rather than strictly “near the end” or an immediate progressive decline from the first tap without rebound. The subsequent rebound suggests fatigue/decrement rather than continuous early-onset progressive decrement required to justify a 3 on amplitude alone.

Interruptions/hesitations count

- No explicit interruptions reported; periods don’t show 3–5 clear breaks or >5 interruptions. This supports $\leq$ slight hesitations.

Overall synthesis

- Speed: slight variability; not clearly mild or moderate slowing.
- Interruptions: not clearly 3–5.
- Amplitude: evident decrement that occurs by mid-sequence and then partially recovers.

Given MDS-UPDRS scoring rules allow assigning the highest applicable criterion among speed, interruptions, or amplitude, the amplitude pattern meeting “decrements midway through the 10-tap sequence” supports a score of Mild (2).

Final score: 2
